

Why did we choose POSIX Message Queues ?

Answer :

We selected POSIX Message Queues because our CPU simulator requires communication between three independent processes using small, structured commands, responses, and log messages. Message Queues preserve message boundaries and allow us to define a common message structure containing the message type, value, address, and data. Compared with pipes, they provide a more natural message-oriented communication model. Compared with shared memory, they avoid the additional synchronization complexity required for shared data. Sockets were unnecessary because our processes communicate locally. Therefore, POSIX Message Queues provide a good balance of simplicity, structure, modularity, and suitability for our simulator.

DESIGN 3 PROCESS ARCHITECTURE

